

Diffusion model architectures

Text-to-Image generation

most slides by Denis Kuznedelev

Ilya Drobyshevskiy
Yandex Research



Outline

00. Image Generation metrics

01. Diffusion model architectures

02. Text-To-Image generation

Image generation metrics

Image quality assessment is a subtle and complicated topic with lots of open questions and research problems.

As an estimate of model performance, research papers report various image generation metrics.



Good or Bad?

Image Generation metrics

FID
(Frechet Inception Distance)

Estimate of “closeness” of generated samples to the target data distribution

Computed on features of pretrained on Imagenet-1k Inception v3 model

Assumes that Inception embeddings are *multivariate normal variables*

Requires many samples (10k and more) for reliable estimate of mean and covariance

$$\text{FID}(\mathcal{N}(\mu, \Sigma), \mathcal{N}(\mu', \Sigma')) = \|\mu' - \mu\|_2^2 + \text{tr}(\Sigma + \Sigma' - 2(\Sigma\Sigma')^{1/2})$$

μ, Σ → Mean and Covariance (real)

μ', Σ' → Mean and Covariance (generated)

Image Generation metrics

CLIP Score

Estimate of “prompt alignment”

Compute cosine similarity for each text/image pair.
Average the per-pair scores over the dataset.

$$\text{CLIP}_{\text{score}} = \frac{(f_{\text{text}}, f_{\text{image}})}{\|f_{\text{text}}\| \|f_{\text{image}}\|}$$

Depends on the choice of specific CLIP text
and vision model

Image Generation metrics

Another approach is to learn human preferences from annotated data

CLIP-like model is fine-tuned on user-preference data

Metric examples: PickScore, ImageReward, HPS, etc

Approximating Human Preferences

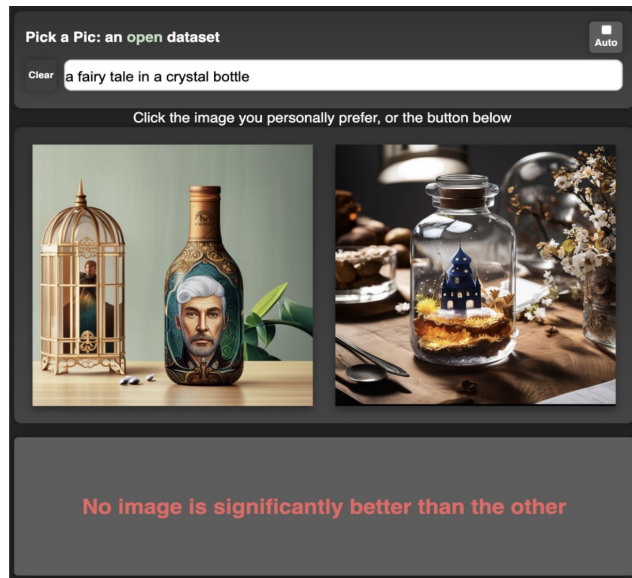


Image Generation metrics

However, **no** automated metric is fully reliable

In practice, **human preference data** is necessary

An image can be evaluated from several aspects

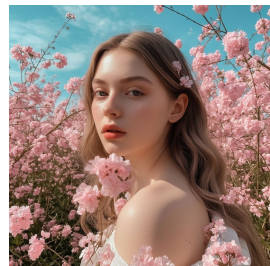
A careful design of instruction provided to human annotators is important

Side-by-Side comparison

Relevance



Aesthetics



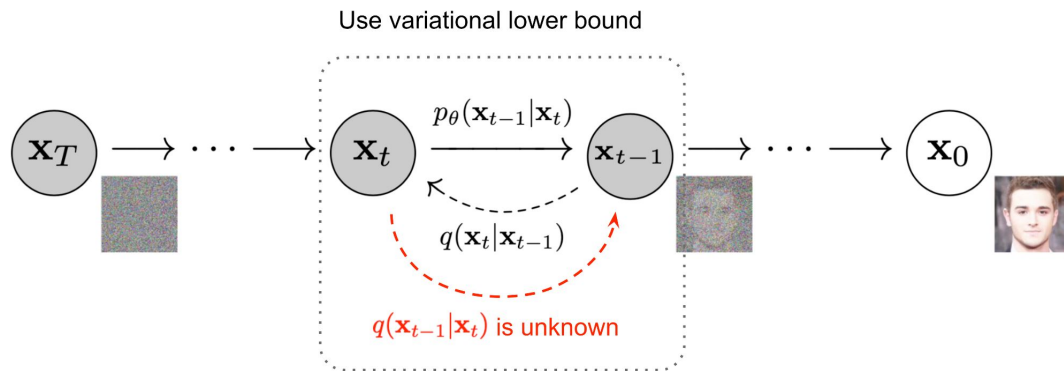
Complexity



Defects



Diffusion model architectures



- Diffusion models transform noise into sample from target data distribution.
- Model outputs are of the same shape (B, H, W, C) as inputs.

Which NN architectures may produce output of the same shape as input?

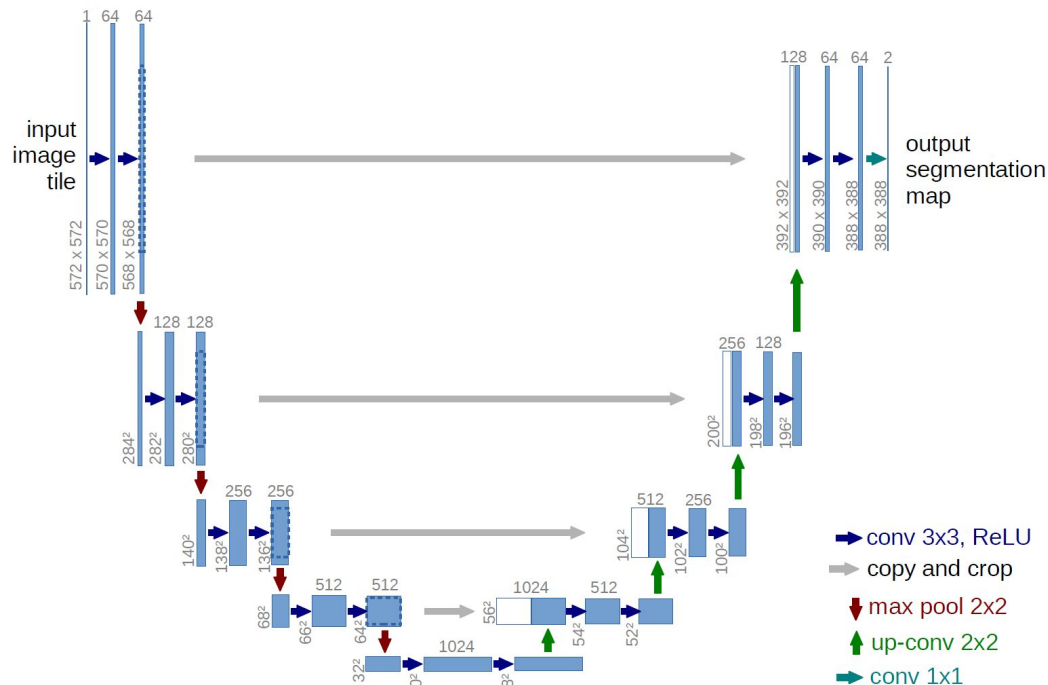
UNet

The most popular architecture is old fellow **UNet**, familiar from the segmentation task

UNet is a stack of Residual Convolutional Blocks operating on different resolutions

Most diffusion models also include Attention blocks

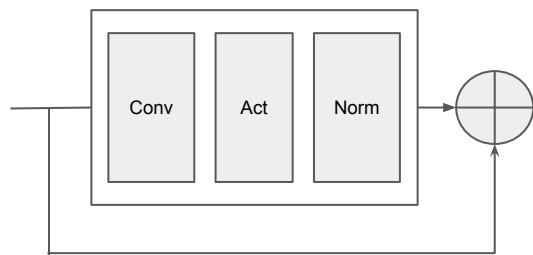
Deeper layers have more channels and operate on smaller features maps



UNet

ResBlock

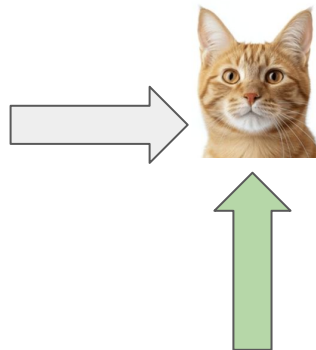
x N



Additive Skip Connection

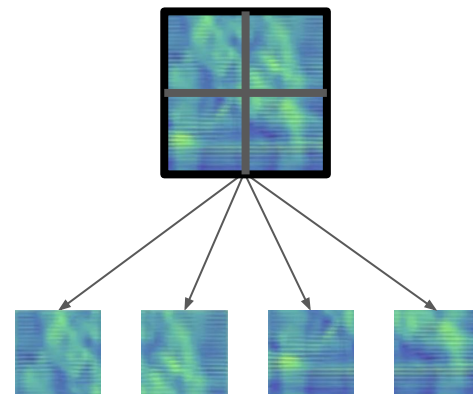
Concatenation
in Decoder

`torch.cat`



Upsampled feature map

Attention



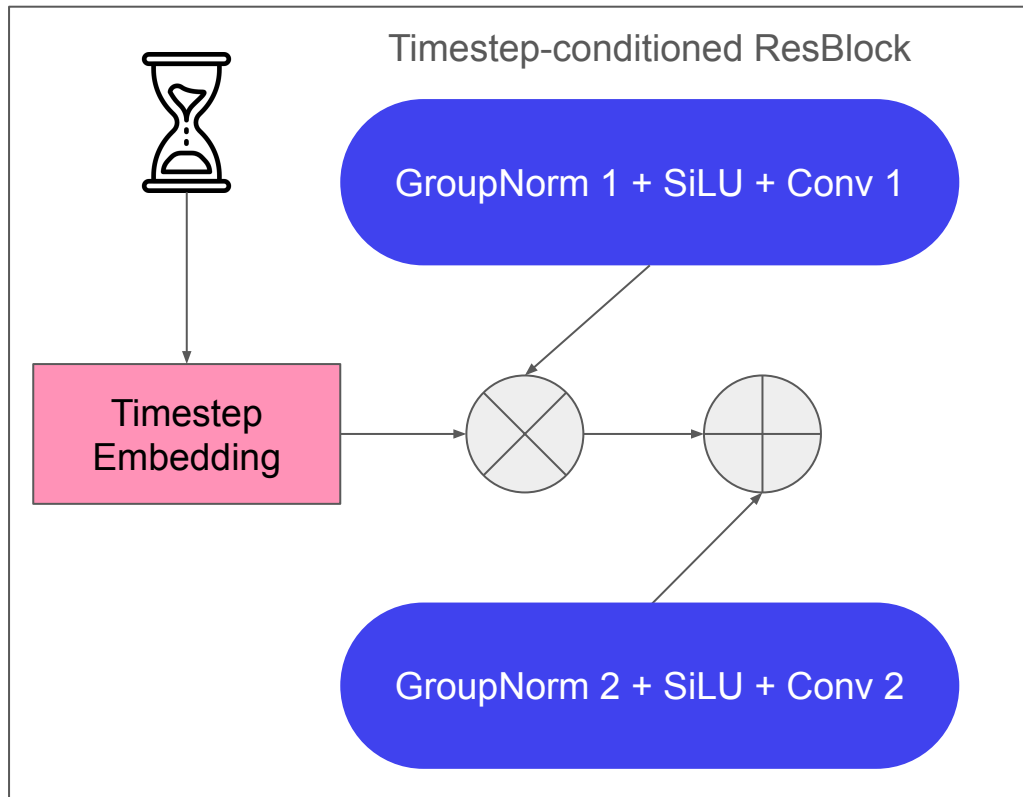
Treat pixels of feature map as
sequence of tokens

UNet

One has to account for the diffusion
timestep conditioning

Standard option is **activation modulation**
in residual blocks

$$y = \text{scale}(t)x + \text{shift}(t)$$



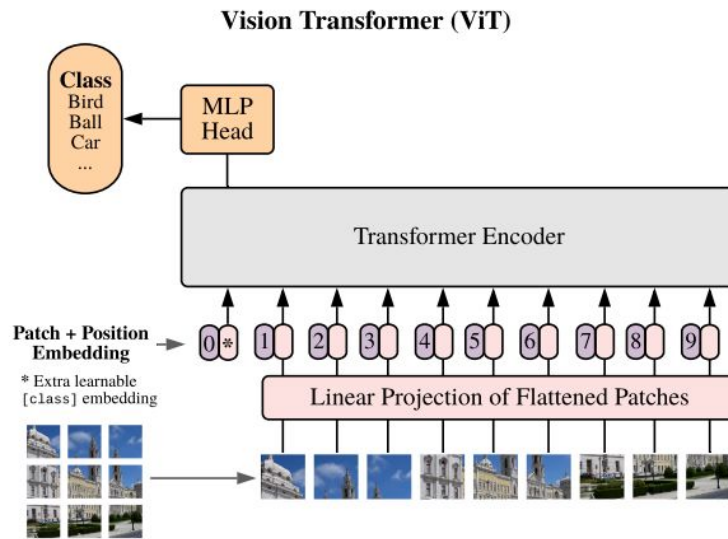
Diffusion Transformer

What about transformer?

A Diffusion model architecture could be built entirely from Transformer blocks

Vision Transformer (ViT) divides the input image into small patches, applies convolution, and processes the sequence like Language Models

Token positions are encoded using 2D positional embeddings



Diffusion Transformer

What about transformer?

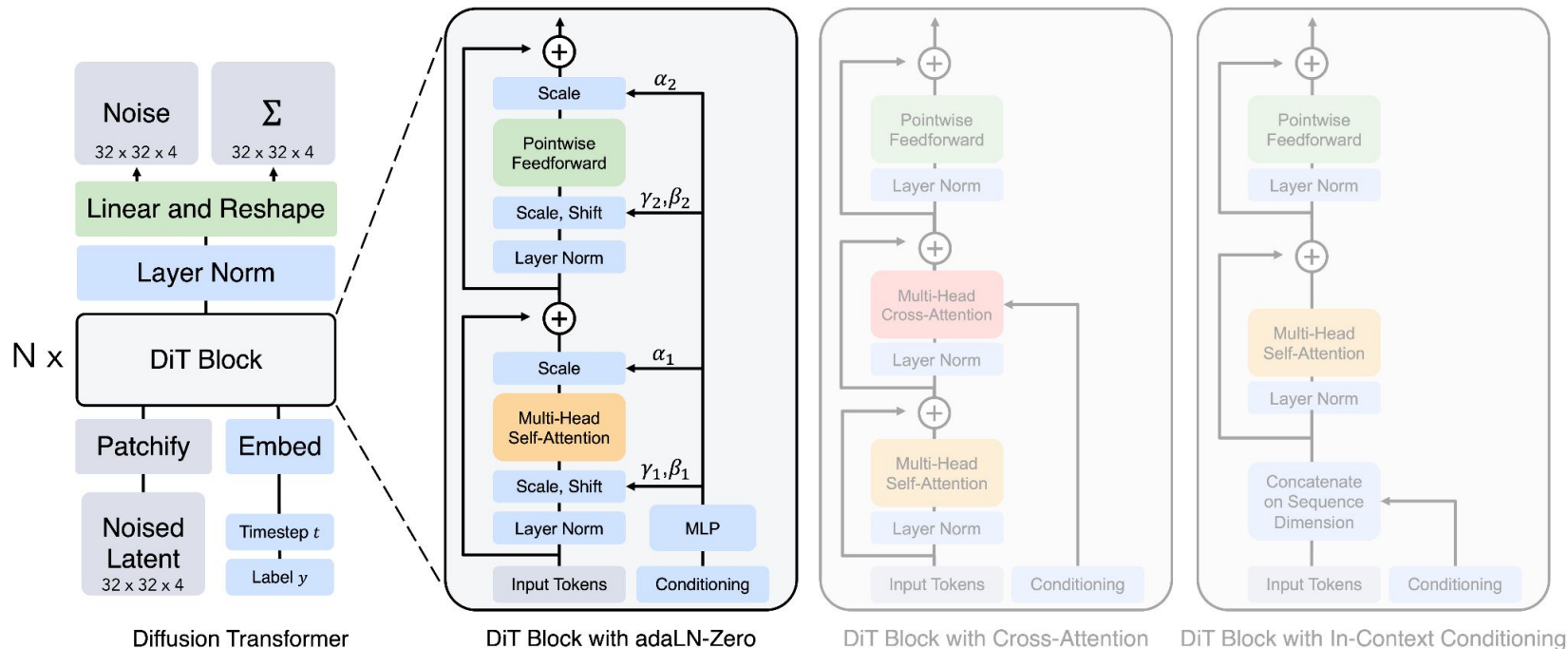
Similarly to ViT one splits image into small patches

Timestep (and class) conditioning is performed via AdaLN

$$\text{AdaLN}(x, c, t) = \text{scale}(c, t) \text{LN}(x) + \text{shift}(c, t)$$

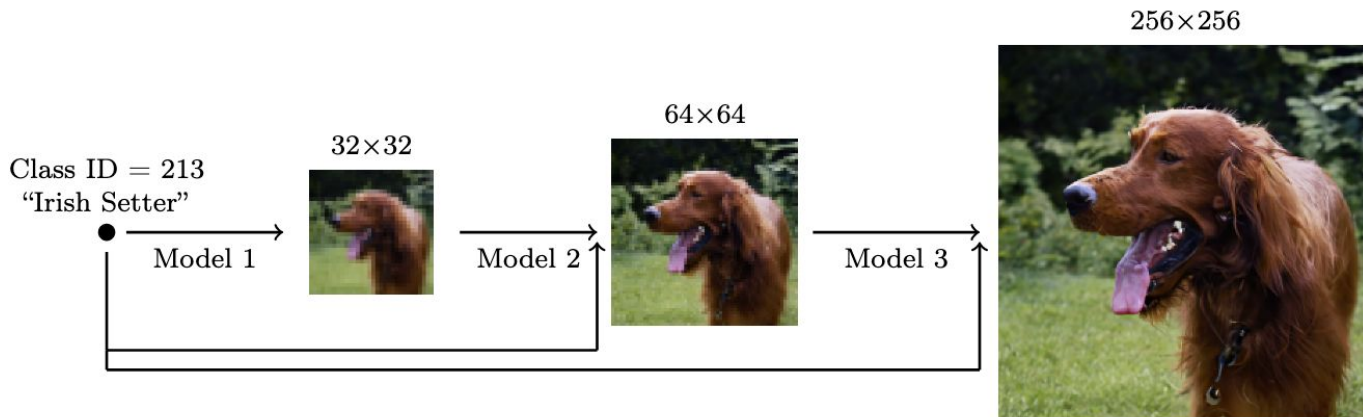
Degrees of freedom are the model size and patch size

Diffusion Transformer



Diffusion model architectures

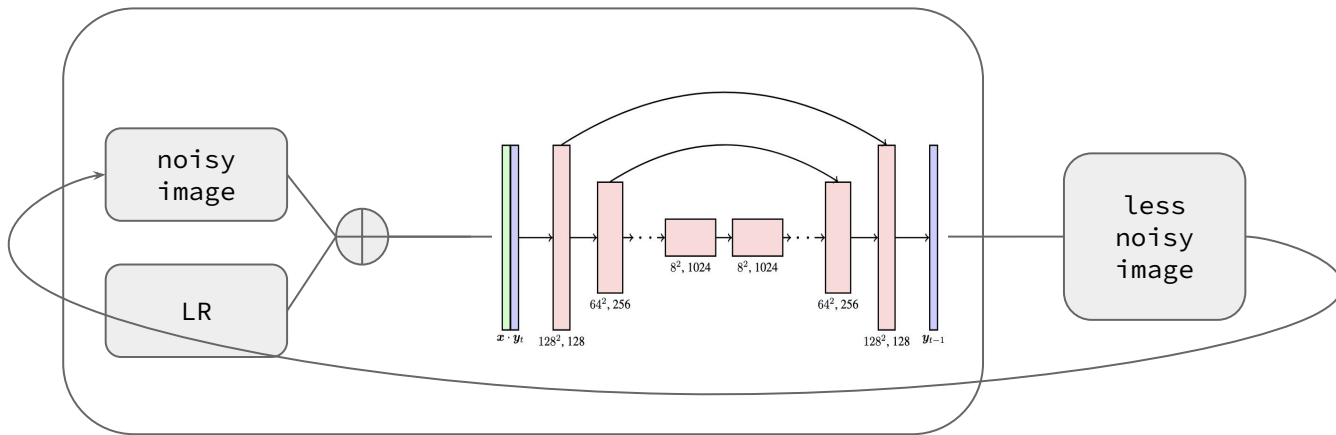
Cascaded Diffusion



Generate low resolution image, and then upscale through one or multiple Super Resolution models

Each Super Resolution model is usually a Diffusion model

Diffusion model architectures



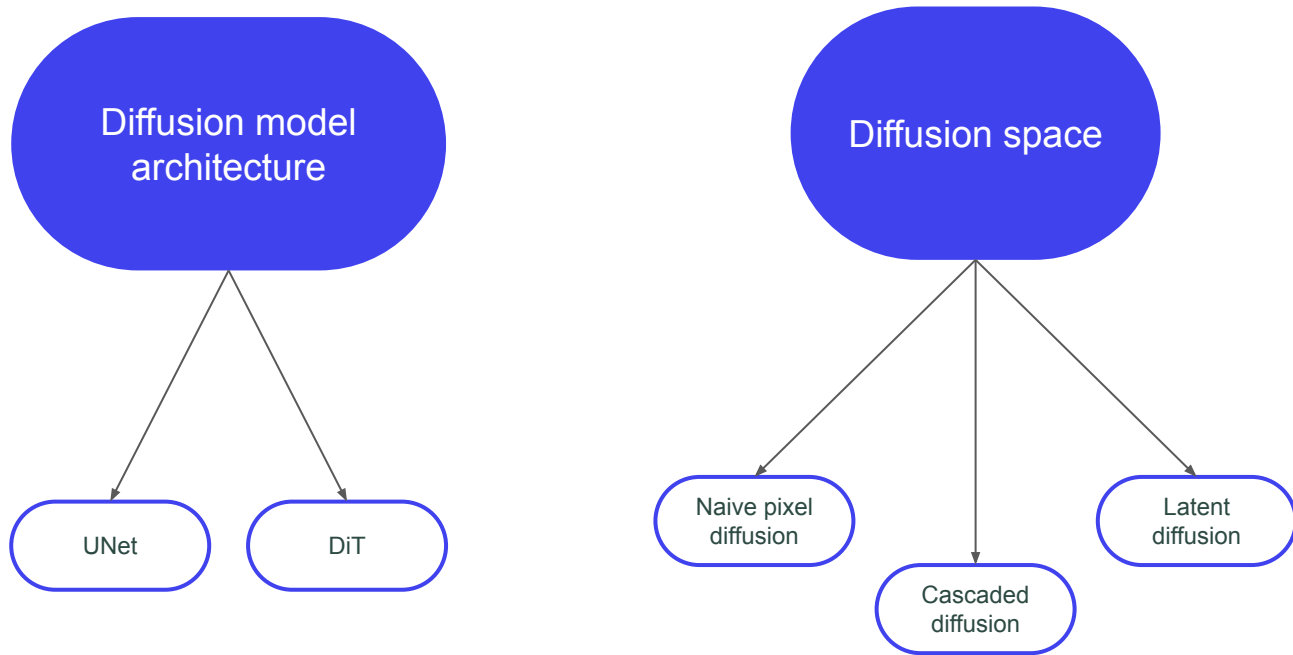
Diffusion SR model - standard UNet for pixel-space diffusion

The main difference is the low-resolution image conditioning via concatenation with input noise

SR models usually have fewer parameters and work well with less sampling steps:

- Base model ~1B parameters, 20-50 sampling steps
- SR model ~300-500M parameters, 10-20 sampling steps

Diffusion model architectures

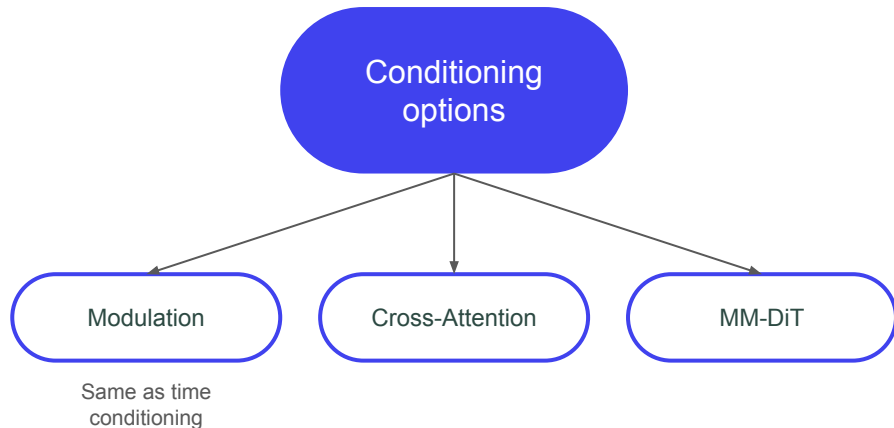


Text2Image generation

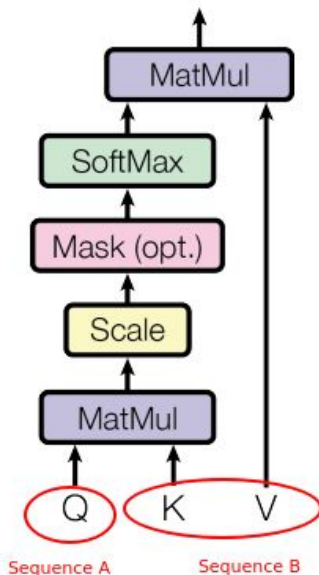
One would like to condition the generation process on the text prompt

LLM and CLIPs can produce decent text embedding

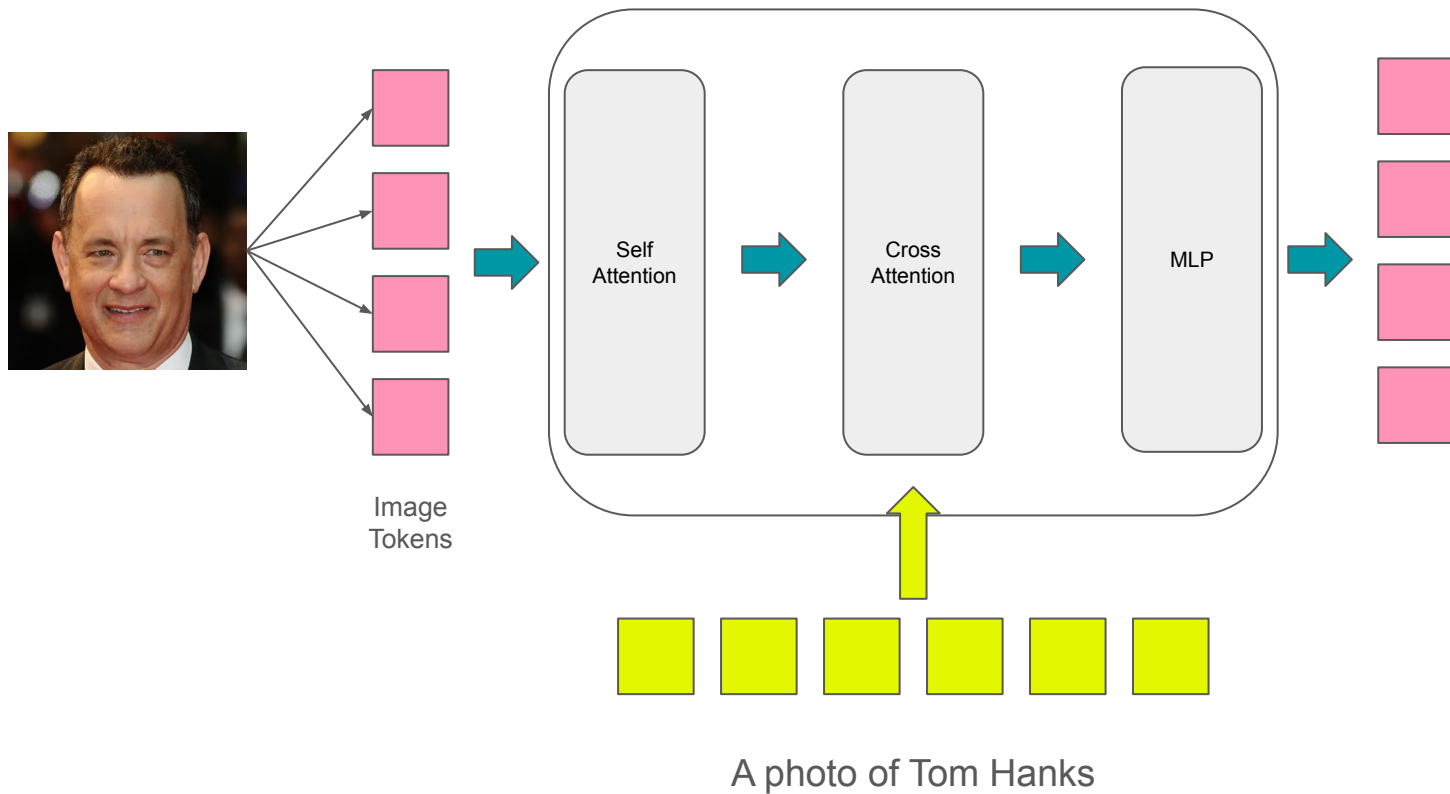
We have a sequence of tokens (vectors) as the output



CROSS ATTENTION



Text2Image generation



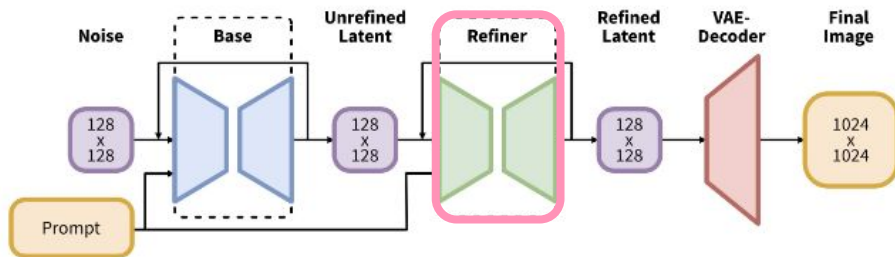
SDXL

2 text encoders (CLIP ViT-L & OpenCLIP ViT-bigG)

Refiner module

Can generate images up to 1024x1024 and rectangular images (4:3, 3:4, 16:9, 9:16)

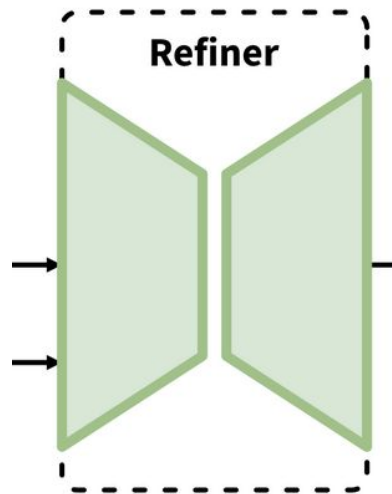
Crop and image size conditioning on training



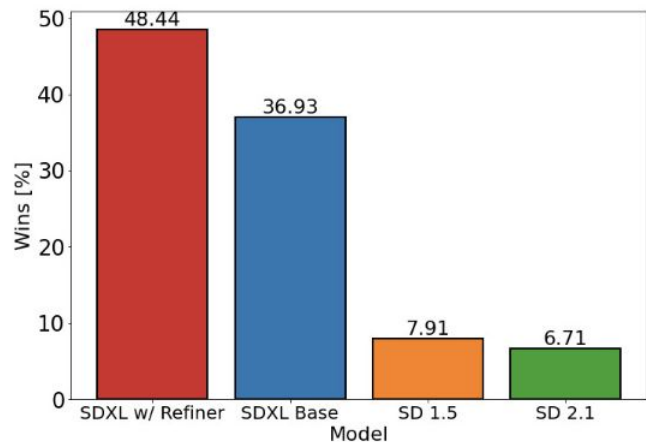
SDXL

Generation produced by the original model is noised back (to relatively small noise level)

Refiner model is another **diffusion model**, conditioned on prompt, running denoising on the final diffusion steps



SDXL



User preference study



Left - w/o Refiner, right - w/ Refiner

Stable Diffusion 3

MM-DiT instead of UNet

8B parameters

3 text encoders (CLIP ViT-L, OpenCLIP ViT-bigG & T5 XXL)

16-channel VAE

QK-normalization in Attention



Prompt: Translucent pig, inside is a smaller pig.



Prompt: A massive alien space ship that is shaped like a pretzel.



Prompt: A kangaroo holding a beer, wearing ski goggles and passionately singing silly songs.



Prompt: An entire universe inside a bottle sitting on the shelf at walmart on sale.



Prompt: A cheeseburger with juicy beef patties and melted cheese sits on top of a toilet that looks like a throne and stands in the middle of the royal chamber.



Prompt: This dreamlike digital art captures a vibrant, kaleidoscopic bird in a lush rainforest



Prompt: A car made out of vegetables.

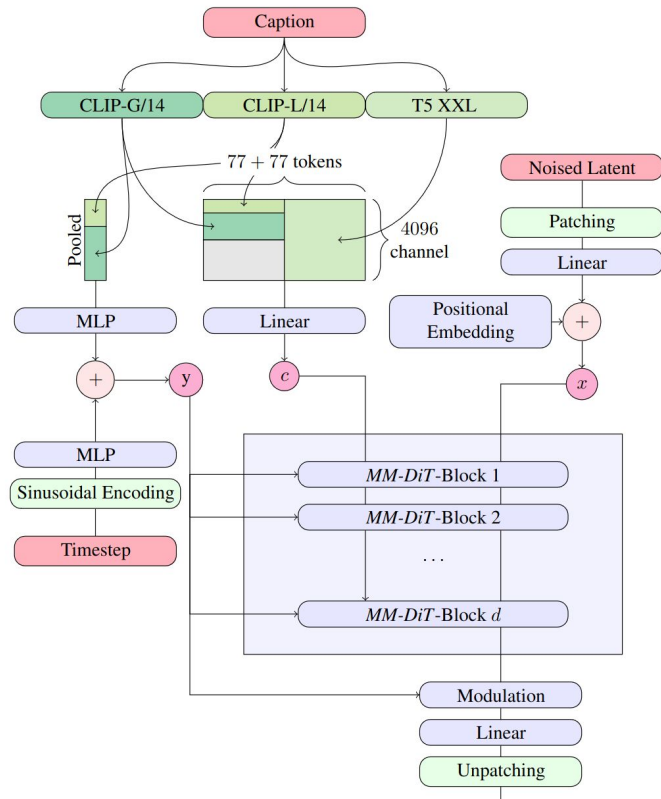


Prompt: Heat death of the universe line art

Stable Diffusion 3

Text embeddings are used as follows:

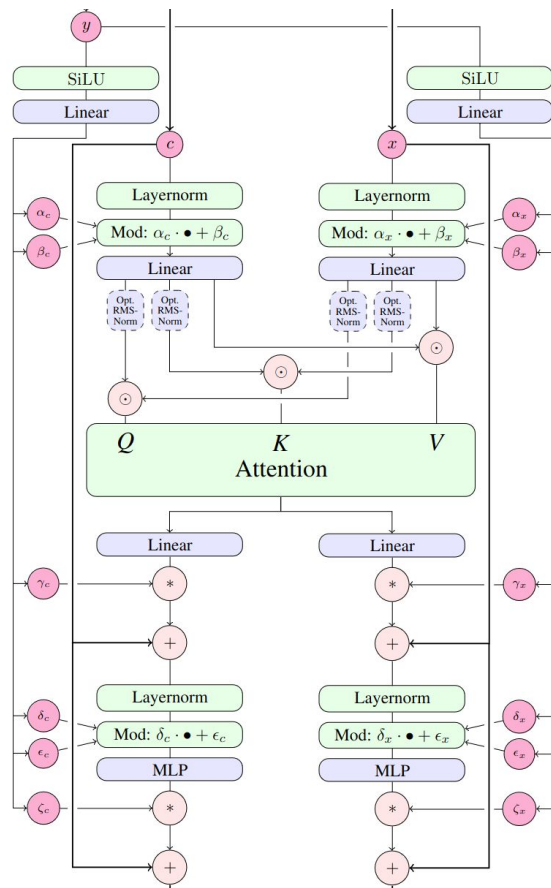
- The pooled CLIP embeddings are added to the timestep embedding and used in AdaLN
- All embeddings are concatenated and fed into the MM-DiT block



Stable Diffusion 3

MM-DiT replaces cross-attention with joint attention over image and text tokens

Before and after attention image and text tokens are transformed as separate sequences



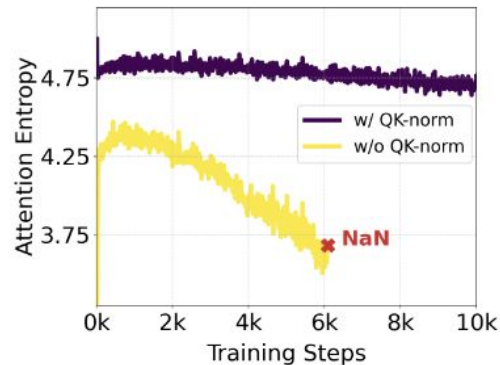
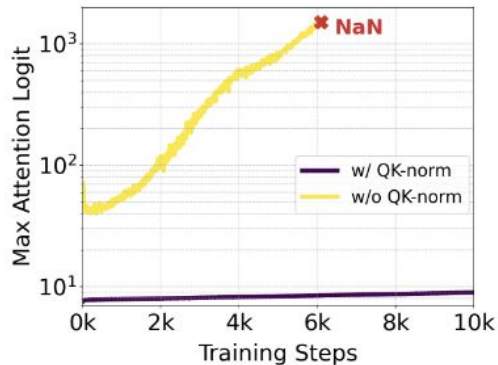
Stable Diffusion 3

SD3 applies RMSNorm with a learned scale to queries and keys before attention.

$$\hat{Q} = \text{RMSNorm}(Q)$$

$$\hat{K} = \text{RMSNorm}(K)$$

$$A = \text{softmax}\left(\frac{\hat{Q}\hat{K}^\top}{\sqrt{d}}\right)$$



QK-normalization is crucial for training stability

FLUX.1

From creators of Stable Diffusion...

MM-DiT with 12B parameters

Two block types:

- Double-stream block (image and text tokens processed as separate sequences but they share the same attention)
- Single-stream block (concatenated tokens are processed together)

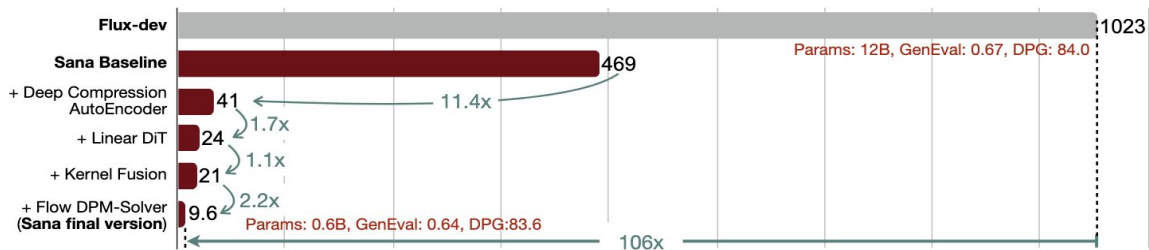
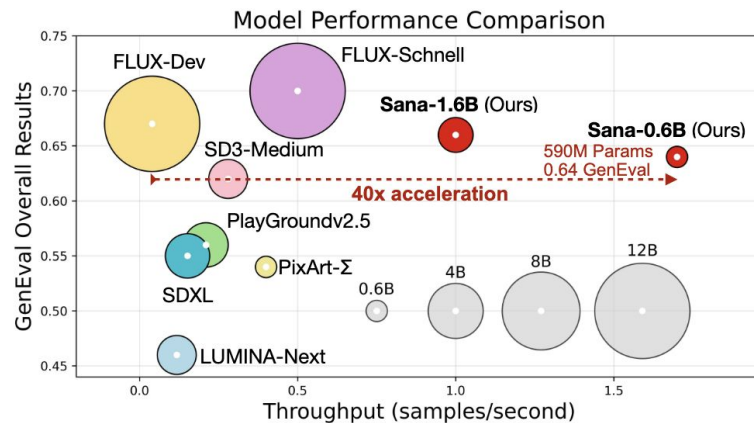


SANA

Adopted Deep Compression Autoencoder (DC-AE)
with high spatial compression ratio (32x)

Replaced standard Attention by Linear Attention

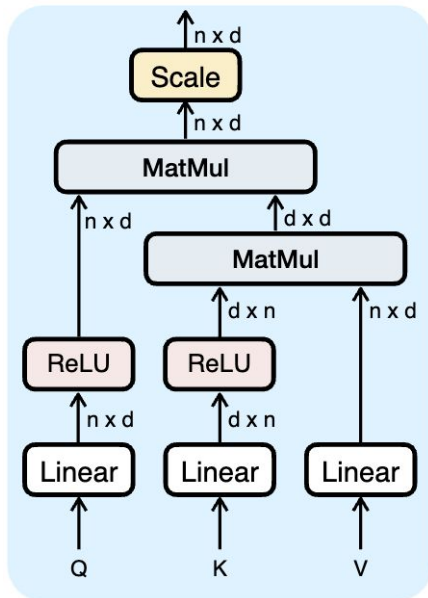
Replaced T5 to Decoder-only Gemma and
removed CLIP



Xie, Enze, et al. "Sana: Efficient high-resolution image synthesis with linear diffusion transformers." *arXiv preprint arXiv:2410.10629* (2024).

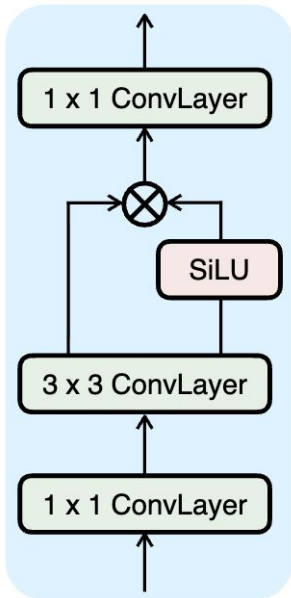
SANA

Linear Attention



Cost : $O(n)$

Mix-FFN



$$\text{ReLU}(QK^T)V \approx \text{ReLU}(Q)\text{ReLU}(K)^T V$$

Modern SOTA

Most SOTA models today use MM-DiT block

Examples:

- FLUX.2 (architecturally the same as FLUX.1 but it replaces CLIP + T5 with Mistral)
- Z-Image

